CHAPTER-1
INTRODUCTION

1.1 Introduction

A hydraulic press machine is a mechanical device that utilizes hydraulic pressure to compress materials into desired shapes and forms. Based on Pascal's Law, which states that pressure applied to a confined fluid is transmitted equally in all directions, the hydraulic press amplifies a small input force into a much larger output force. This principle enables it to perform heavy-duty operations such as moulding, forging, punching, deep drawing, and metal forming with high efficiency and precision. Hydraulic presses are widely used in various industries, including automotive, aerospace, construction, and manufacturing, due to their versatility, reliability, and ability to generate consistent force. With advancements in automation and control systems, modern hydraulic presses offer enhanced safety, energy efficiency, and accuracy, making them indispensable equipment in industrial applications.

In the rapidly progressing world of industrial automation and manufacturing, machines that can deliver high force with precision and reliability are essential. Among such machinery, the hydraulic press machine plays a pivotal role in shaping, moulding, pressing, and compressing materials in a wide variety of industrial applications. Based on the fundamental principle of fluid mechanics, hydraulic presses have emerged as a dominant solution for producing high-pressure force in controlled environments. They are widely utilized in industries like automotive, aerospace, metal forming, plastic moulding, powder metallurgy, and even in laboratory material testing.

At its core, a hydraulic press is a mechanical device that utilizes the properties of fluids under pressure to generate a compressive force. The operating principle of the hydraulic press is based on Pascal's Law, which states that "when pressure is applied to a confined fluid, the pressure is transmitted equally and undiminished in all directions." This allows hydraulic presses to produce extremely large output forces using relatively small input forces. For example, with just a small-diameter plunger applying force to a fluid, the pressure created can push a much larger ram or piston, enabling operations such as forming, stamping, punching, deep drawing, and compacting with great efficiency and control.

Hydraulic presses have a simple but robust structure, generally composed of key components such as a hydraulic cylinder, plunger or ram, pump, valves, frame, and control system. The press frame can come in different configurations such as H-frame, C-frame, or four-pillar depending on the requirements of the operation. The hydraulic pump is used to generate fluid pressure, which is directed through valves to move the cylinder. The press ram then exerts force on the workpiece positioned on the die, thereby completing the pressing operation.

Powder metallurgy (PM) is a progressive and widely used manufacturing process that enables the production of high-strength, complex-shaped components with minimal material waste. It involves three primary steps: powder production, compaction, and sintering. Among these, the compaction process plays a crucial role in determining the mechanical integrity and dimensional precision of the final product. In this step, metal powders are compressed within a die under high pressure to form a "green compact," which is later sintered to achieve the desired mechanical properties.

Hydraulic presses are widely used for the compaction stage in powder metallurgy due to their ability to apply consistent, controllable, and high-pressure forces. These presses ensure uniform density distribution throughout the compacted part, which is essential for maintaining structural performance. However, commercially available hydraulic presses for PM applications are often expensive and may not be tailored for specific production needs, especially in small-scale industries, educational institutions, or research laboratories.

The aim of this project is to design and develop a cost-effective, reliable, and user-friendly hydraulic press machine specifically for powder metallurgy applications. The design focuses on achieving sufficient compaction pressure, ease of operation, and adaptability to different powder materials and component geometries. Key components of the system include a hydraulic cylinder, pump, pressure gauge, die-punch setup, and a supporting frame, all of which are carefully selected and assembled to withstand the required loads and provide precise control during operation.

This project not only contributes to making powder metallurgy processes more accessible and economical but also serves as a foundation for further development in automated and semi-automated compaction systems. The designed press is tested under various load conditions and material types to validate its performance and reliability.

Powder metallurgy (PM) is a well-established and increasingly important manufacturing technology that allows the production of components with complex geometries, controlled porosity, and high material utilization. This process has become vital in industries such as automotive, aerospace, electronics, and biomedical engineering due to its advantages in cost-efficiency, material conservation, and performance enhancement. The fundamental steps in powder metallurgy include powder preparation, mixing, compaction, and sintering. Among these, compaction is particularly critical, as it directly influences the density, strength, and dimensional accuracy of the final sintered component.

The design and development of a hydraulic press machine for powder metallurgy represents a significant technological advancement in modern manufacturing processes, particularly in the realm of precision engineering and materials science. Powder metallurgy (PM) is a technique used to produce complex and high-performance components from powdered raw materials, offering benefits such as material efficiency, precise control over composition, and reduced machining requirements. The hydraulic press plays a critical role in this process by compacting metal powders into desired shapes with high pressure before sintering. In developing such a machine, engineering considerations span a wide range—from structural integrity and force.

generation to system automation, thermal management, and ergonomic design. The hydraulic system must be capable of generating consistent and controlled pressure, typically ranging from a few tons to several hundred tons, depending on the application. The press must ensure uniform density distribution and avoid defects such as cracks, voids, or non-uniform hardness, which can compromise the final product's performance. Furthermore, the machine must be designed with operational safety, energy efficiency, ease of maintenance, and adaptability to various die sets and powder materials. The integration of CAD modeling, finite element analysis (FEA), and advanced control systems enables precise design validation and performance optimization. Ultimately, the successful development of a hydraulic press tailored for powder metallurgy not only enhances the quality and reliability of sintered components but also supports cost-effective, high-volume manufacturing in industries like automotive, aerospace, biomedical, and tool-making sectors.

the design and development of a hydraulic press machine for powder metallurgy is a multidisciplinary engineering challenge that involves the integration of mechanical design, fluid power systems, materials science, and manufacturing technologies. Powder metallurgy is gaining increasing prominence due to its ability to fabricate near-net shape components with minimal material waste and excellent mechanical properties. The hydraulic press used in this process must be meticulously designed to withstand high loads, ensure repeatability, and achieve the required compaction force necessary to form green compacts with consistent density and dimensional accuracy.

In the conceptualization phase, the focus is placed on identifying the load requirements, press capacity (usually measured in tons), and structural configuration—typically H-frame or C-frame—based on the type of product and production volume. Structural analysis, often performed using finite element methods, is essential to ensure the frame and platen can endure repeated stress without deformation. The selection of hydraulic components such as cylinders, pumps, valves, and control units is equally critical, as these dictate the performance, response time, and precision of the press. In powder metallurgy, it is crucial to maintain a uniform pressure distribution across the die cavity, as uneven compaction can lead to internal flaws, density gradients, or deformation during sintering. Hence, the press mechanism must be designed to offer high accuracy in motion control, facilitated by programmable logic controllers (PLCs), pressure sensors, and feedback systems.

Moreover, the development process takes into account the integration of features such as die alignment systems, ejector mechanisms for easy removal of the compact, and safety interlocks to protect operators. The system may also include heating elements for warm compaction processes, depending on the powder material. The development of such a machine also demands careful selection of materials for critical components, ensuring wear resistance, fatigue strength, and long-term durability. Energy efficiency and environmental sustainability are additional considerations, with modern presses incorporating variable displacement pumps or servo-hydraulic systems to reduce energy consumption during idle phases.

1.2 OBJECTIVES

- To design a hydraulic press machine for the compaction of metal powders.
- Study and selection of oil and material.
- Design and calculation of cylinder
- To Validate the Design Using HyperMesh Simulation

CHAPTER-2

LITERATURE REVIEW

[1] Tory Robison et al... Powder metallurgy compacting presses usually are mechanically or hydraulically driven, but they can incorporate a combination of mechanically, hydraulically, and pneumatically driven systems. This article provides a comparison of mechanical and hydraulic presses based on the cost, production rate, and machine overload protection. The article lists the classification of powder metallurgy parts based on complexity of shapes as suggested by the Metal Powder Industries Federation, such as Class I parts, Class II parts, Class III parts, and Class IV parts. It describes rigid tooling compaction and details the powder-fill ratio considerations for these classes. The article elaborates on the types of tooling systems and presses used for these classes. Some important factors and components used in designing a tool are also described. Finally, the article considers tool materials, including punches, core rods, and punch clamp rings.

[2] Guoyuan Wen et al…The copper-based powder metallurgy friction material, modified with dual ceramic components (WC/B_4C), underwent friction experiments paired with custom-made carbon ceramic discs. The overall findings are as follows: The judicious addition of dual ceramic components (WC/B_4C) can enhance the mechanical, thermal, and frictional performance of copper-based powder metallurgy friction materials. When the WC/B_4C ratio is 5:3, the material exhibits the optimal mechanical, thermal, and frictional properties. B_4C and WC, after braking, form friction films of B_2O_3 and WO_3. Additionally, the friction film of copper-based powder metallurgy friction materials can be categorized into an oxide layer, a deformation layer, and a matrix layer. The oxide layer primarily consists of Fe_2O_3, the deformation layer is mainly composed of WO_3 and B_2O_3, and the matrix layer is primarily composed of CuO and Cu_2O. With the increase in the WC/B_4C ratio, the primary wear mechanisms of copper-based powder metallurgy friction materials shift from abrasive wear to mild fatigue wear, mild fatigue wear progressing to severe fatigue wear, adhesive wear, and fatigue wear, accompanied by a gradual intensification of oxidative wear.

[3] Rajesh Purohit et al…The present paper deals with the development of Ni-Ti shape memory alloys through a novel powder metallurgy process. The Ni-Ti shape memory alloy is widely used in automobile, aerospace, sensor, vibration, actuators, biomedical and other applications. The Ni-Ti shape memory alloys mixed in the weight percent of their atomic number. The powders were then cold compacted in a hydraulic press at pressure of 300 MPa and subsequently sintered at 1150°C for 30 minutes at 25 MPa pressure in vacuum (10^{-5}Torr). The microstructure and effect of rolling is to done in sintered and hot-pressed sample. Further the rolling of hot-pressed sample was done to improve the mechanical properties and testing of shape memory effect on Ni-Ti strip was investigated. The rolled thin strip was further used for testing of mechanical properties and shape memory behavior. The hot-pressed sample rolling in thin strip. The effect of the rolling on the hot-pressed sample was marginal on the mechanical properties and transformation temperature of Ni-Ti shape memory alloys.

[4] Tamalet M et al… A detailed analysis of energy consumption throughout the PM production route has been conducted, based on both theoretical and experimental considerations, with the focus on PM parts production. Comparison of the values obtained with actual values from production plants provides useful insights into process development and priorities for energy conservation. Furnace processes (powder annealing and component sintering) are found to dominate energy consumption, hence operations that achieve efficient furnace loading can enjoy significant benefits. Energy input per kilogram of production was found to vary widely among the plants surveyed, depending on part complexity and size, production volume and material.

[5] P. Samal et al... Powder metallurgy compacting presses usually are mechanically or hydraulically driven, but they can incorporate a combination of mechanically, hydraulically, and pneumatically driven systems. This article provides a comparison of mechanical and hydraulic presses based on the cost, production rate, and machine overload protection. The article lists the classification of powder metallurgy parts based on complexity of shapes as suggested by the Metal Powder Industries Federation, such as Class I parts, Class II parts, Class III parts, and Class IV parts. It describes rigid tooling compaction and details the powder-fill ratio considerations for these classes. The article elaborates on the types of tooling systems and presses used for these classes. Some important factors and components used in designing a tool are also described. Finally, the article considers tool materials, including punches, core rods, and punch clamp rings.

[6] M.N. Subramanian et al… Hydraulic Presses in Powder Metallurgy Applications Subramanian conducted a fundamental study on the role of hydraulic presses in powder metallurgy and emphasized the need for a machine that can consistently deliver uniform pressure over the powder compaction surface, which is critical to achieving homogenous density and high green strength of the compacted parts. The study highlighted that unlike mechanical presses, hydraulic presses offer precise control over pressure application, making them more suitable for the powder metallurgy process where material flow and compaction uniformity are of utmost importance. He further elaborated on the design considerations such as cylinder dimensions, piston stroke, die alignment, and the need for programmable logic controllers (PLCs) to automate pressing cycles and enhance production efficiency.

[7] M. Ramesh et al... Design and Fabrication of Hydraulic Press Ramesh and his team designed and fabricated a basic hydraulic press machine intended for small-scale manufacturing processes and demonstrated that the selection of the frame type, particularly the H-type frame, offers superior load distribution and structural rigidity compared to other configurations like the C-frame. Their study meticulously detailed the calculation of cylinder bore, pressure requirements, selection of pump capacity, and fabrication of the frame using mild steel materials. The authors also performed load testing on the fabricated press, showing that the design could successfully withstand the expected operational loads and that the machine offered a cost-effective and structurally reliable alternative for low- to medium-capacity powder compaction applications.

[8] S. Basavaraj et al… Design and Development of a Hydraulic Press In this research, Basavaraj and co-authors aimed at designing and developing a 10-ton hydraulic press for educational and light industrial purposes, placing strong emphasis on frame design, hydraulic cylinder construction, and selection of seal materials. The study included the use of FEA (Finite Element Analysis) to validate the stress-bearing capacity of the press frame, ensuring that the design could withstand operational loads without failure. They concluded that a properly designed press not only enhances durability and safety but also plays a pivotal role in achieving better densification during the compaction of metal powders, which is essential in powder metallurgy**.**

[9] R. Patel et al… Analysis and Design of Hydraulic Press Patel and Shah carried out an extensive analysis on the structural design of a hydraulic press machine, focusing on the load-bearing components such as the main frame, base, and press plates, and conducted stress and strain analysis using ANSYS software. Their results confirmed that careful attention to the thickness of plates, weld joints, and alignment of components significantly impacts the machine's performance, especially when used in precision-based applications like powder metallurgy. They also underlined the importance of optimizing the press structure to avoid excessive deflections and stress concentrations, which could lead to uneven compaction and part defects.

[10] R. Prakash et al… Automation of Hydraulic Press Using PLC This study by Prakash and Sharma explored the integration of automation technologies into the operation of hydraulic presses for powder compaction, highlighting the role of PLCs in increasing process reliability and repeatability. By introducing sensors, limit switches, and timers into the control circuit, the press could automatically manage the stroke, dwell time, and return movement, thereby reducing operator dependence and ensuring consistency in the pressing process. They argued that automation is essential for modern powder metallurgy applications, where part-to-part consistency is necessary, and manual operation introduces variability and error.

[11] A. K. Srivastava et al… Design Optimization of Powder Compacting Hydraulic Press Srivastava focused on optimizing the hydraulic press design to reduce energy consumption while improving operational efficiency, especially for high-volume powder compaction. The paper discussed strategies like using lightweight but high-strength materials for structural components, designing energy-efficient hydraulic circuits, and employing variable displacement pumps. The optimization techniques outlined in the study led to a noticeable reduction in operational costs and improved the life of wear-prone parts such as seals and valves, which are crucial in maintaining the pressure integrity during the pressing cycle.

[12] V. Kumar et al… Material Selection for Hydraulic Cylinder Components. This research provided a comparative analysis of different materials used in the construction of hydraulic cylinders, which are a core component in any press machine. Kumar and his team investigated materials such as EN8 steel, alloy steels, and mild steels under simulated operational conditions involving high-pressure loads, frequent cycling, and thermal variation. Their findings indicated that EN8 steel offers the best combination of mechanical strength, machinability, and fatigue

resistance, making it ideal for applications requiring high-pressure and continuous operation, such as in powder metallurgy where repeated pressing is common.

[13] Singh et al… Performance Evaluation of Hydraulic and Mechanical Presses Singh and Ali conducted a performance comparison between hydraulic and mechanical presses in the context of powder metallurgy, and their experimental evaluation demonstrated that hydraulic presses produced compacts with significantly better uniformity and density. The paper explained that hydraulic systems offer flexibility in pressure control, stroke speed, and dwell time, which are essential for achieving defect-free, high-strength green parts. They also concluded that while mechanical presses might offer higher speed, their limited control mechanisms make them less suitable for the precision needs of the PM industry.

[14] Arora et al… FEA of Hydraulic Press Frame Arora and colleagues used Finite Element Analysis (FEA) to simulate the load behaviour and stress distribution in a 5-ton hydraulic press frame. The simulations revealed critical areas where stress concentrations were likely to occur under maximum loading, and based on these findings, the authors recommended the addition of support ribs and the use of gusset plates in the design. The enhanced frame design was then validated through physical testing, confirming that such improvements not only prevent structural deformation but also help maintain alignment and compaction accuracy in powder metallurgy applications.

[15] D. Patil et al… IoT-Enabled Hydraulic Press for Powder Compaction In this recent study, Patil and his team explored the integration of Internet of Things (IoT) technology with hydraulic press machines used for powder compaction, aiming to create a smart system capable of real-time monitoring and data collection. They implemented pressure sensors, stroke measurement devices, and a cloud-based dashboard that allowed users to track machine performance remotely. This innovation was particularly useful in predictive maintenance, where early detection of component wear and hydraulic leakage could prevent costly breakdowns. Their research represents a major step towards Industry 4.0 compliance in powder metallurgy manufacturing environments.

[16] S.K. Sahu et al… – "Design and Development of Hydraulic Press for Industrial Applications" This study focuses on the conceptual and structural design of a hydraulic press intended for industrial manufacturing, including powder metallurgy applications. The researchers developed a compact 20-ton press system using finite element analysis to test the structural reliability of the H-frame. Emphasis was placed on stress distribution, selection of appropriate hydraulic cylinders, and optimizing pump size for energy efficiency. The study concluded that the mechanical integrity and precision of hydraulic movement were essential for applications involving powder compaction, where uniform pressure distribution directly affects the quality of the green compact.

[17] Singh et al… – "Design and Structural Analysis of Hydraulic Press for PM Applications" This study focused on designing a 10-ton capacity hydraulic press intended specifically for powder metallurgy components. Using SolidWorks for modeling and ANSYS for finite element analysis (FEA), the authors evaluated the stress and deformation characteristics of the frame, platens, and cylinder mounts under maximum load. Their results emphasized the importance of optimizing the frame geometry (typically H-type) to reduce bending stress concentrations. They also addressed the need for robust weld joints and suggested using mild steel with high yield strength to resist fatigue loading during repeated pressing cycles. The paper concluded that careful structural design and simulation are critical to ensuring safety, longevity, and product accuracy in PM presses.

[18] Prasad et al… – "Multi-Level Powder Compaction Using Custom Hydraulic Press Machine" This study tackled the issue of forming complex parts with variable thickness and density zones by using multi-level compaction techniques. The authors designed a hydraulic press with multiple synchronized rams to allow different pressure profiles during compaction. The results showed better density distribution and improved sintered properties in parts like stepped gears and structural brackets. The paper argued for custom press designs tailored to part geometry rather than using a single-cylinder approach.

[19] Dr. P. S. Kulkarni et al… – "Automation in Hydraulic Press Machines for Precision Manufacturing" This paper addresses the need for automation in hydraulic presses used for powder metallurgy. The researcher emphasized using electronic pressure sensors, stroke counters, and human-machine interface (HMI) panels to improve control over the compaction process. With fine-tuned feedback loops, the press could respond to variations in powder volume or compaction resistance in real-time, thus maintaining uniform density. The paper also reviewed challenges in synchronizing the upper and lower platen during dual-action compaction to avoid shear cracks in the compact.

[20] Zhao et al… – "Advanced Powder Compaction Techniques using High-Tonnage Hydraulic Presses" This international study explored how advanced hydraulic systems—especially servo-hydraulic drives—are revolutionizing powder compaction. It reviewed innovations in press design where digital pressure control and position sensing provide micron-level accuracy. These presses are typically used in the manufacturing of hard-metal tools, automotive gear parts, and precision ceramics. The study highlighted that while initial costs are high, the productivity gains and improved mechanical properties of sintered parts.

Based on the detailed review of existing literature, the present work focuses on the **design and development of a hydraulic press specifically suited for powder metallurgy (PM) applications**. From previous studies, it is evident that hydraulic presses offer superior **pressure control, uniform compaction, flexibility, and automation capability** compared to mechanical presses, which are critical for achieving defect-free green compacts. Overall, the developed hydraulic press demonstrates improved **compaction uniformity, safety, and adaptability**, aligning with modern PM requirements and providing a practical solution for efficient powder compaction.

CHAPTER-3

METHODOLOGY

3.1 FLOW CHART

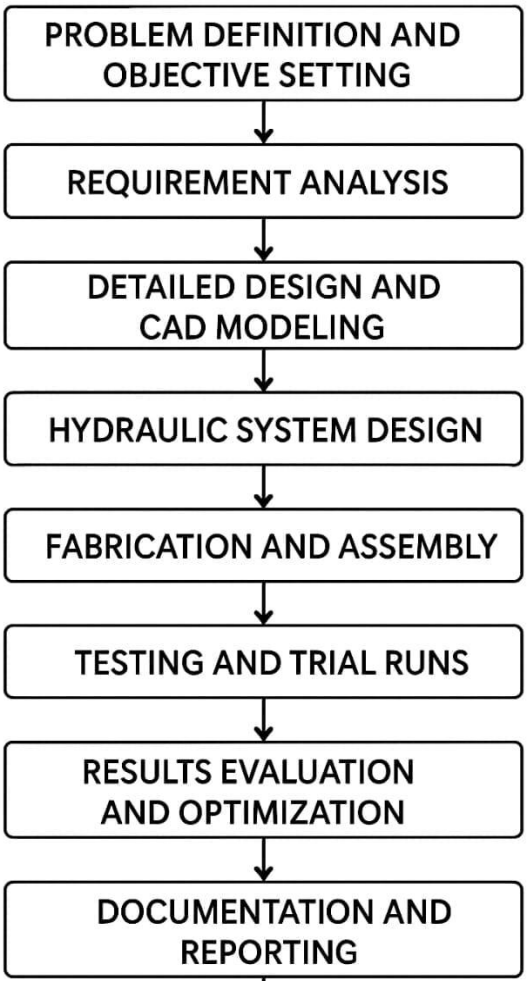


Fig 3.1 Flow Chart

3.2 Problem Definition and Objective Setting

Objective: To design and develop a cost-effective, efficient, and reliable hydraulic press machine suitable for powder metallurgy applications.

- Scope:
    - Compacting metal powders into precise shapes.
    - Ensuring uniform pressure distribution.
    - Enhancing production capacity and dimensional accuracy

3.3 Requirement Analysis

- Identify the material properties (metal powders: iron, copper, etc.).
- Define the required pressing force (typically 10–250 tons depending on application).
- Determine die and punch dimensions, press stroke length, and cycle time.
- Set accuracy and density targets for green compacts**.**

3.4 Conceptual Design

- Choose the type of press:
    - C-frame, H-frame, or Four-post press based on load and accessibility.
- Sketch rough concepts for:
    - Frame structure
    - Hydraulic cylinder and piston assembly
    - Die and punch arrangement
    - Control systems

3.5 Detailed Design and CAD Modeling

- Use CAD software (e.g., SolidWorks, AutoCAD, or CREO) for 3D modeling.
- Design components:
    - Hydraulic cylinder (bore, stroke, pressure rating)
    - Press bed and platen
    - Guide columns and supports
    - Die assembly
- Material selection (typically mild steel or cast iron for structure).

3.6 Hydraulic System Design

- Select key components:
    - Hydraulic pump (gear, vane, or piston)
    - Direction control valve
    - Pressure relief valve
    - Flow control valve
    - Hydraulic reservoir and filter

3.7 Analysis and Validation

- Perform Finite Element Analysis (FEA) on:
    - Frame to check stress and deformation under load.
    - Platen and die under pressing force.
- Validate:
    - Structural integrity
    - Factor of safety
    - Deformation limits

3.8 Fabrication and Assembly

- Procure raw materials and standard parts.
- Fabricate:
    - Frame, base plate, and cylinder housing.
- Assemble:
    - Hydraulic cylinder with seals
    - Valve and pipe fittings
    - Mounting of die and punch

3.9 Testing and Trial Runs

- Conduct dry runs to check alignment, stroke, and pressure build-up.
- Perform actual powder compaction using metal powder:
- Measure green compact dimensions, weight, and density.
- Evaluate:
    - Repeatability
    - Cycle time
    - Surface finish and integrity of compacted parts

3.1.1 Results Evaluation and Optimization

- Compare actual outcomes to design goals.
- Modify:
    - Hydraulic pressure or punch speed
    - Die tolerances
- Implement design improvements based on trial results

3.1.2 Documentation and Reporting

- Prepare detailed reports including:
    - Design calculations
    - CAD models
    - FEA results
    - Hydraulic circuit
    - Testing data
    - Cost analysis

CHAPTER-4

DESIGN & CALCULATION

4.1-2D Drawing of Hydraulic Press Machine

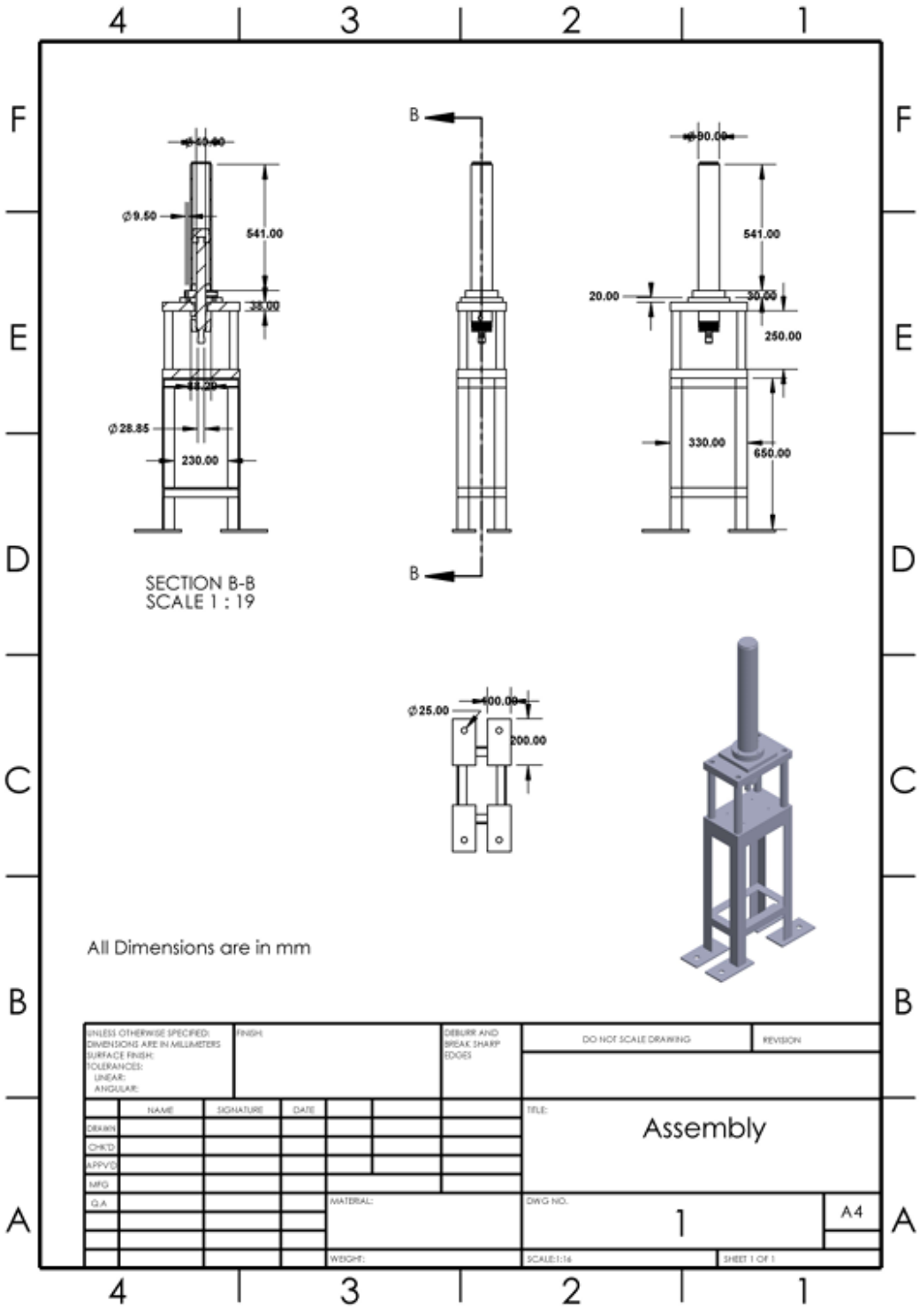


Fig 4.1.1 - 2D Drawing of Hydraulic Press Machine Assembly

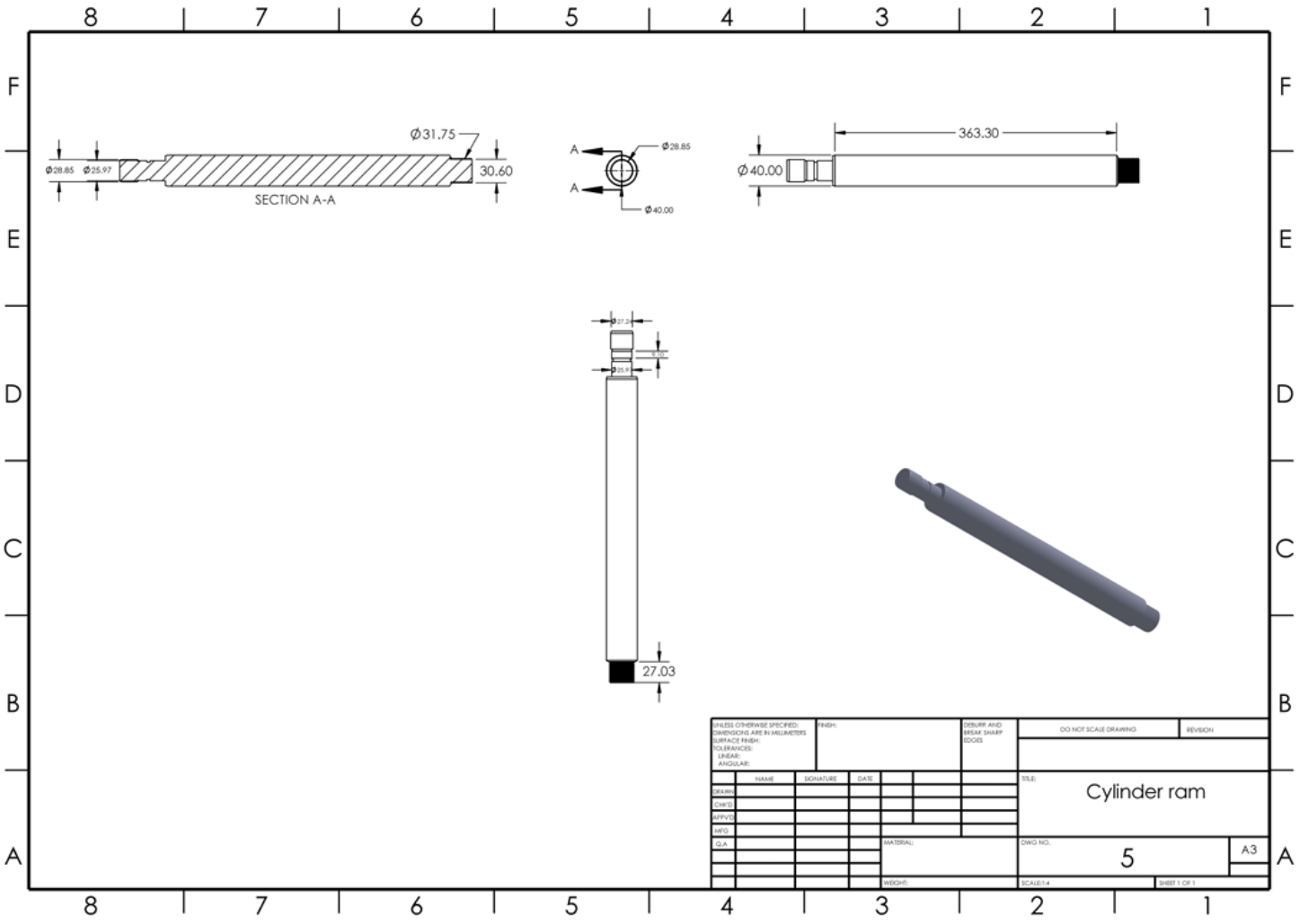


Fig 4.1.2 – 2D Drawing of Cylinder Ram

Table 4.1.2 Dimensional Specifications of Cylinder Ram

Sl. No.	Description	Dimension (mm)
1	Overall length of cylinder ram	363.30
2	Main body diameter	Ø 40.00
3	Reduced end diameter	Ø 31.75
4	Length of reduced end	30.60
5	Step diameter at rod end (1)	Ø 28.85
6	Step diameter at rod end (2)	Ø 25.97
7	End face thickness	27.03
8	Outer diameter of end portion	Ø 28.85
9	Inner diameter at end	Ø 40.00

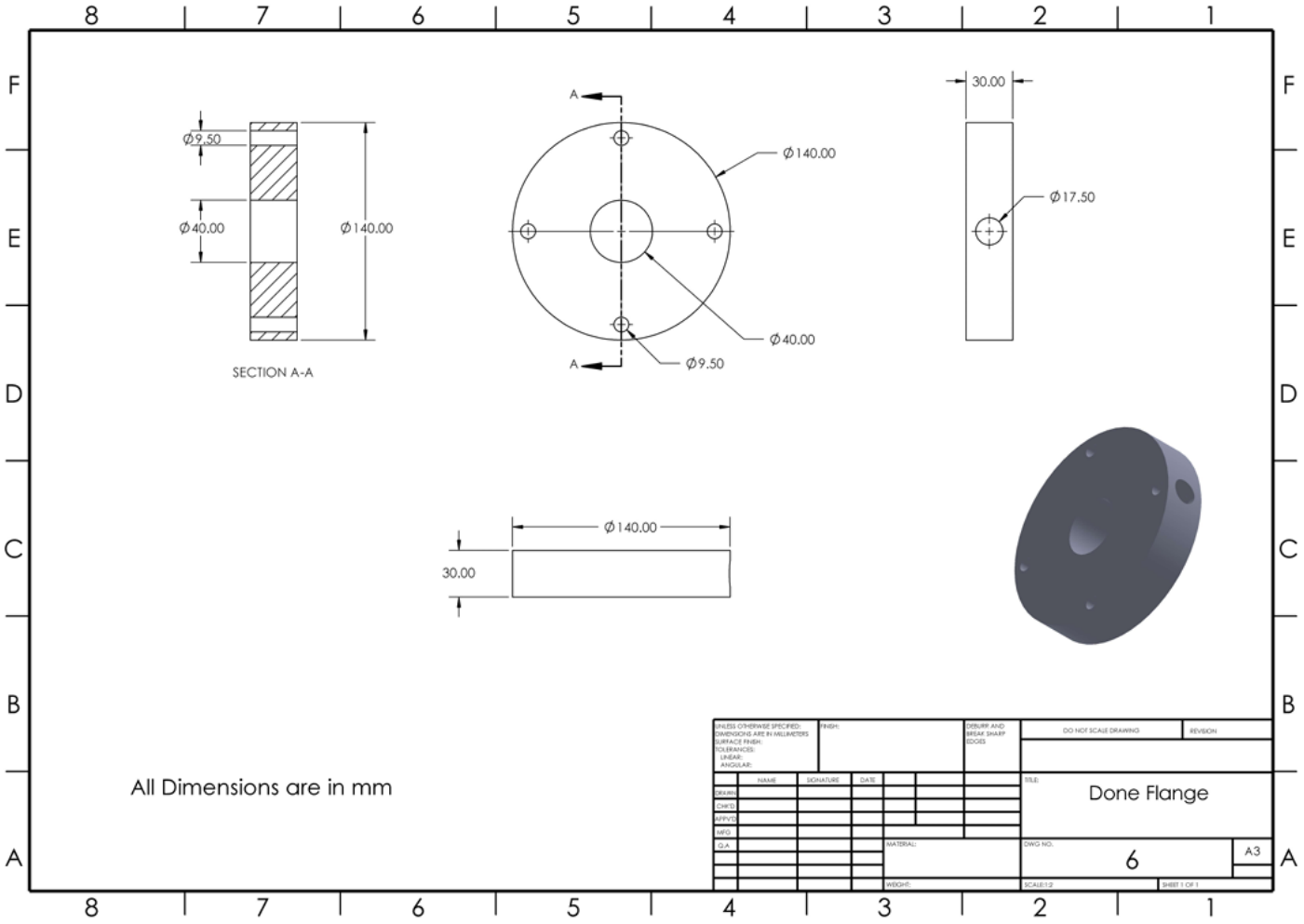


Fig 4.1.3 – 2D Drawing of Flange

Table 4.1.3 Dimensional Specifications of Flange

Sl. No.	Description	Dimension (mm)
1	Outer diameter of flange	Ø140.00
2	Thickness of flange	30.00
3	Central bore diameter	Ø40.00
4	Bolt hole diameter	Ø9.50
5	Number of bolt holes	4
6	Side hole diameter	Ø17.50
7	Overall width in side view	30.00
8	Section thickness (A–A)	30.00
9	Bolt hole pitch circle diameter	Ø140.00

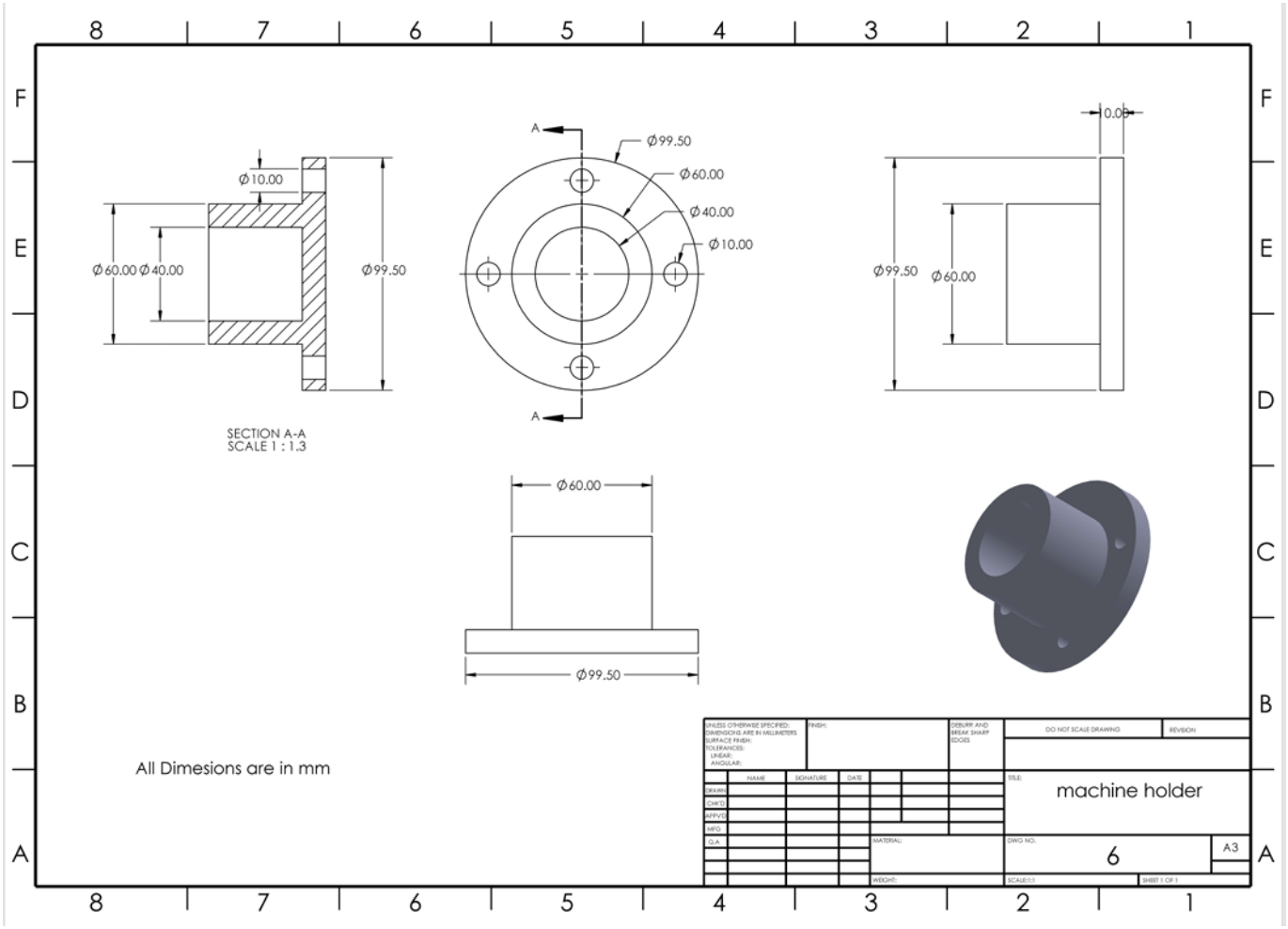


Fig 4.1.4 – 2D Drawing of Cylinder Holder

Table 4.1.4 Dimensional Specifications of Cylinder Holder

Sr. No.	Description	Dimension (mm)
1	Outer flange diameter	Ø99.50
2	Flange thickness	10.00
3	Outer diameter of holder body	Ø60.00
4	Inner bore diameter	Ø40.00
5	Height of holder body	60.00
6	Bolt hole diameter	Ø10.00
7	Number of bolt holes	4
8	Bolt hole pitch circle diameter	Ø99.50
9	Overall diameter (front view)	Ø99.50

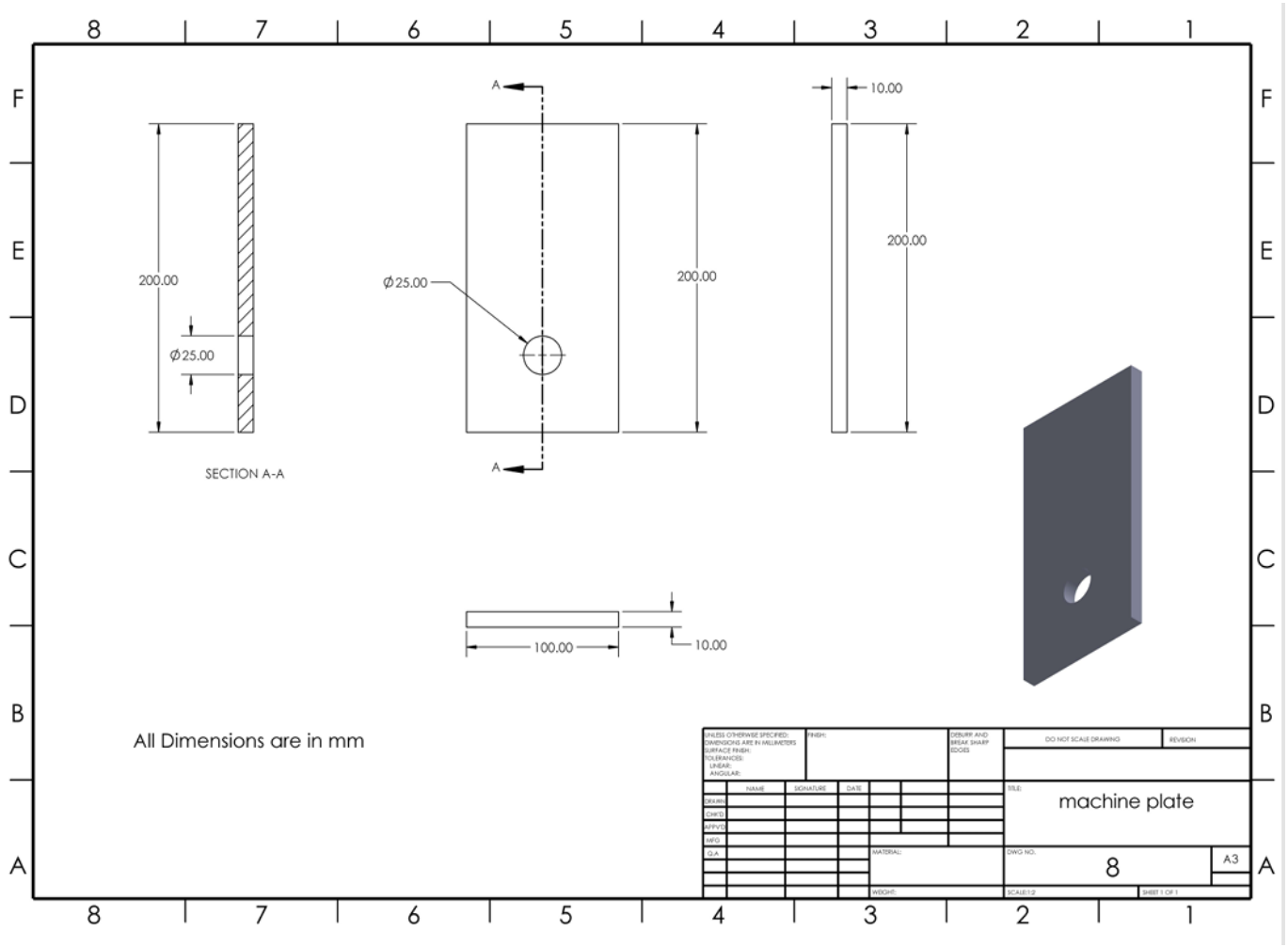


Fig 4.1.5 – 2D Drawing of Machine Plate

Table 4.1.5 Dimensional Specifications of Machine Plate

Sr. No.	Description	Dimension (mm)
1	Overall height of plate	200.00
2	Overall width of plate	100.00
3	Thickness of plate	10.00
4	Diameter of central hole	Ø25.00
5	Vertical distance of hole	200.00
6	Plate height (side view)	200.00
7	Plate thickness (side view)	10.00
8	Section A–A height	200.00
9	Section A–A hole diameter	Ø25.00

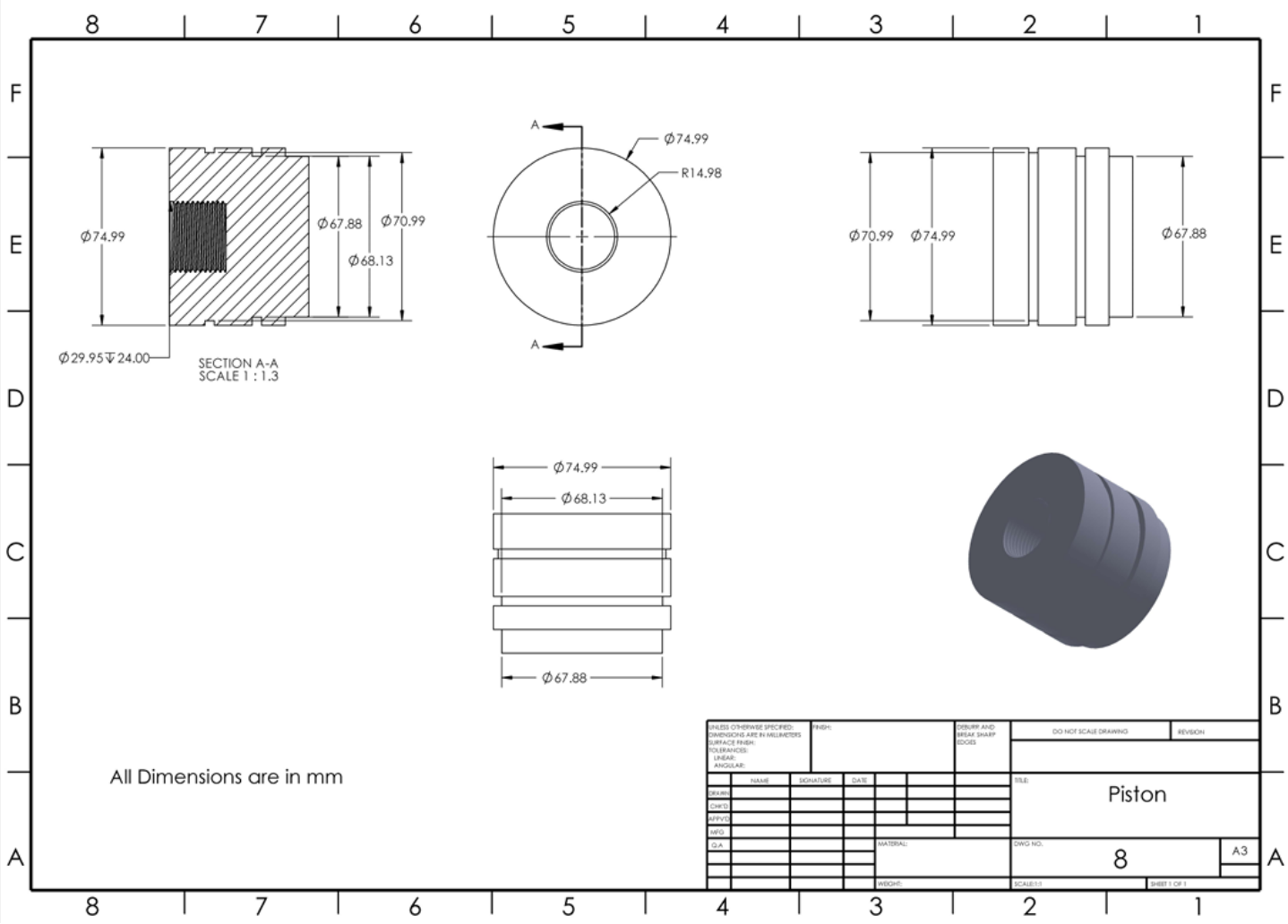
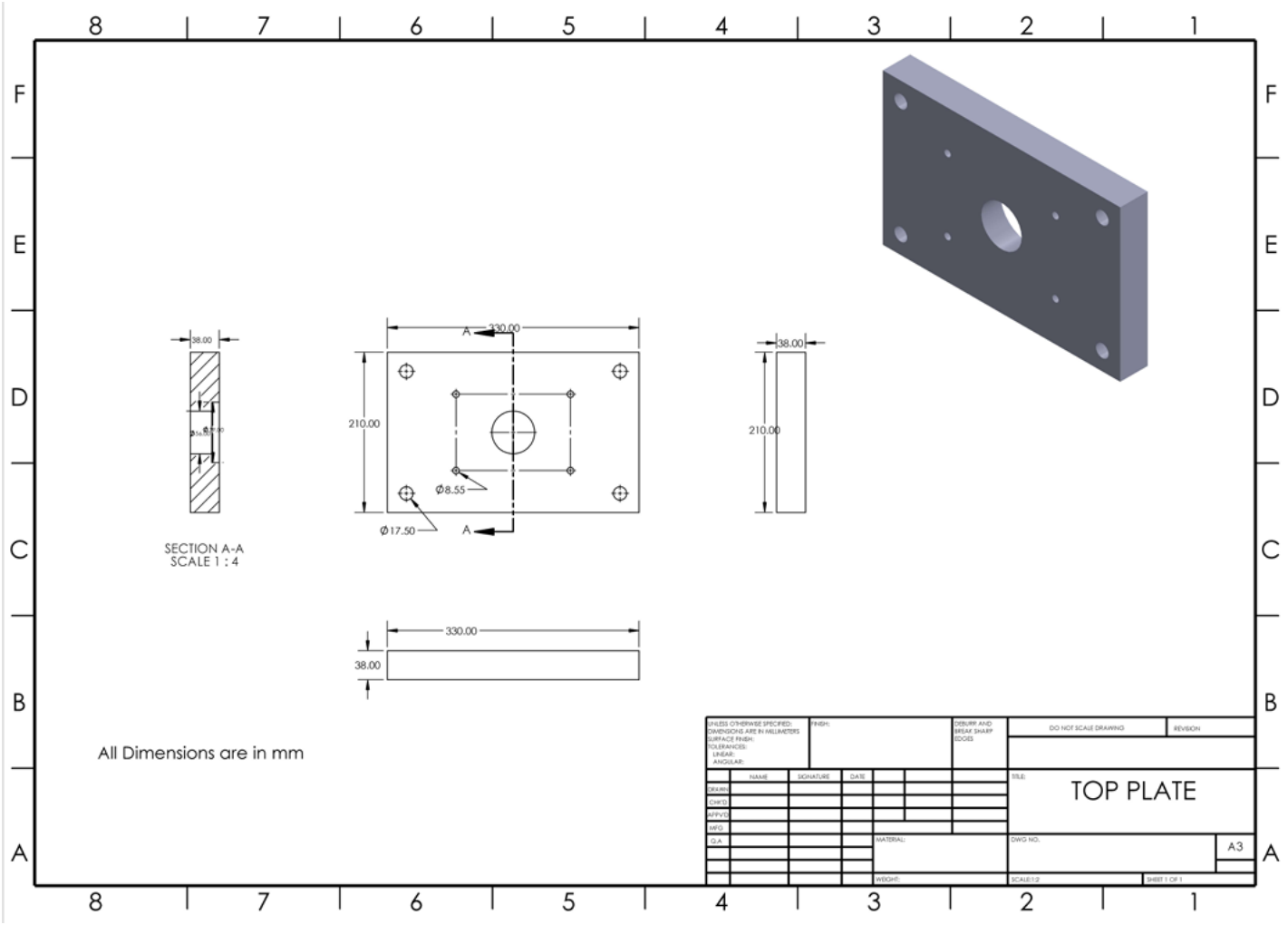


Fig 4.1.6 – 2D Drawing of Piston

Table 4.1.6 Dimensional Specifications of Piston

Sl. No.	Description	Dimension (mm)
1	Outer diameter of piston (maximum)	Ø74.99
2	Outer diameter – middle step	Ø70.99
3	Outer diameter – lower step	Ø67.88
4	Inner bore diameter	Ø29.95
5	Inner bore step diameter	Ø24.00
6	Intermediate diameter	Ø68.13
7	Radius at edge	R14.98
8	Bottom outer diameter	Ø67.88
9	Top outer diameter	Ø74.99



4.1.7 – 2D Drawing of Top Plate

Table 4.1.7 Dimensional Specifications of Top Plate

Sl. No.	Description	Dimension (mm)
1	Overall length of top plate	330.00
2	Overall width of top plate	210.00
3	Thickness of top plate	38.00
4	Diameter of central hole	Ø85.55
5	Corner hole diameter	Ø17.50
6	Number of corner holes	4
7	Plate length (side view)	330.00
8	Plate width (side view)	210.00
9	Plate thickness (section A–A)	38.00

4.2 – 3D Model of Hydraulic Press Machine

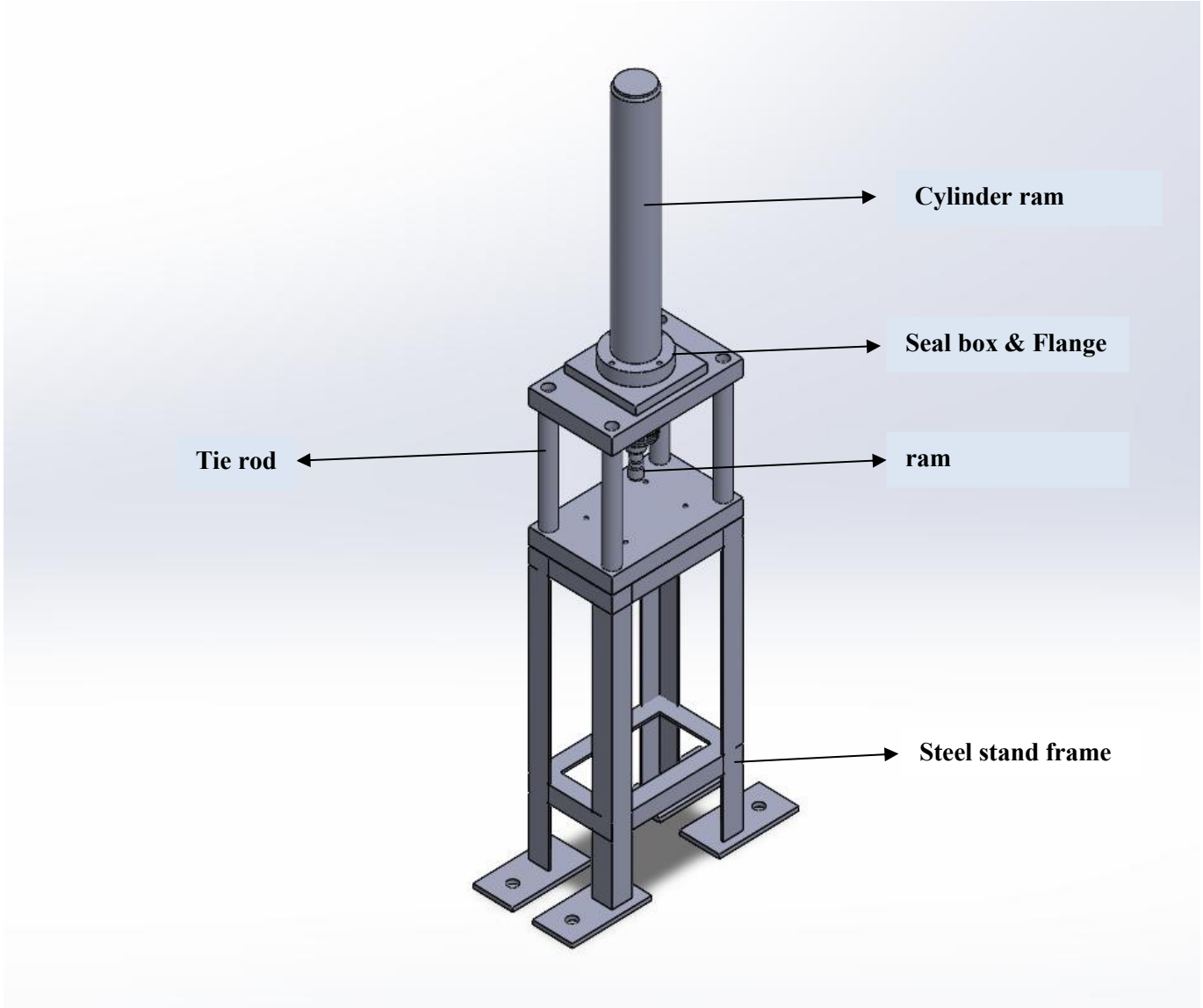


Fig 4.2.1 – 3D Model of Hydraulic Press Machine

4.3 Software Part

4.3.1 SolidWorks

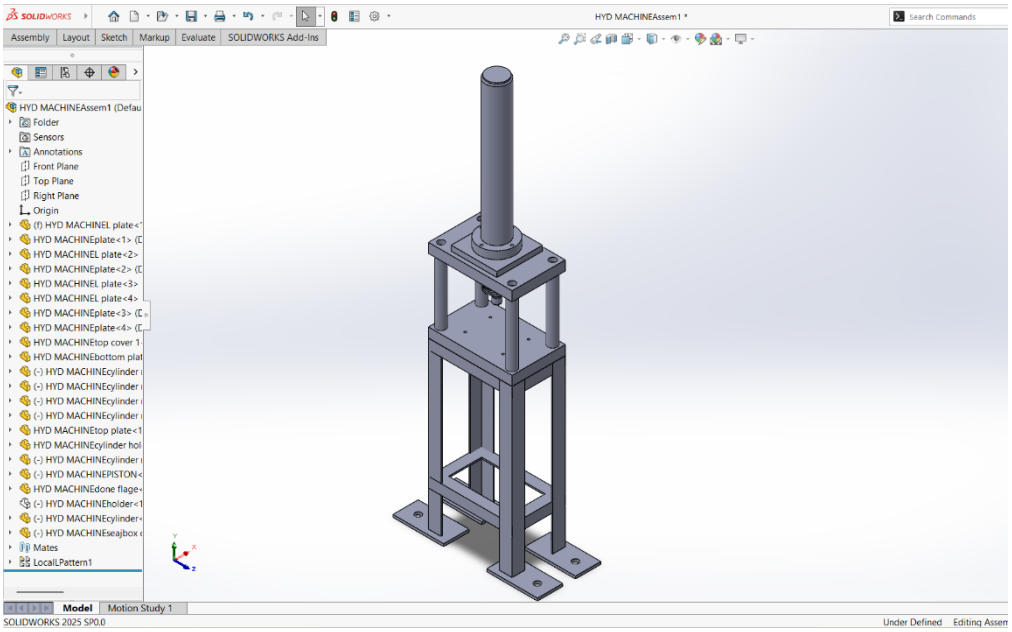


Fig 4.3.1 – SolidWorks

SolidWorks is a powerful CAD software ideal for designing mechanical projects. It allows you to create detailed 3D models, simulate real-world conditions, and optimize designs for performance. For thermal-related projects, SolidWorks offers tools like Flow Simulation and Thermal Studies to analyze heat transfer and material behavior. You can validate designs, detect interferences, and ensure functionality through motion analysis. It also helps prepare 2D drawings for manufacturing and supports efficient collaboration with your team. Overall, it's a versatile tool for turning ideas into precise, functional designs.

4.4 Design Calculations

4.4.1 Basic Parameters

- Pressing Force (F): 3 tons = 29,430 N
- Working Pressure (P): 100 bar = 10 MPa
- Ram (Piston Rod) Diameter = 36mm
- Working Pressure= 100 bar

➢ **Effective Piston Area**

$A = {}^{\pi}/_{4} \times D^2$

$A = {}^{\pi}/_{4} \times 63^2$

$A = 3.117mm^2$

- **Working Pressure**

$$P = {Load}/{Area}$$
$$P = {3000}/{3.117}$$
$$= 9624\ kg$$
$$= {9624}/{100}$$
$$P = 96.24 \approx 100bar$$

- **Calculation of hole cylinder**

63mm bore → working pressure = 100 bar
For safer side we will increases bar by 30%
100×1.3
130 bars
31.17 $cm^2 \times 130\ {kg}/{cm^2}$
=40552.1 kg
=4. 0521 ton

- **Ram selection**

In MS steel $\frac{4052.3\ KG}{850\ {kg}/{cm^2}}$ =4.768 area (850 is the standard of Ms tensile stress)

To find Dia $\frac{area}{{\pi}/{4}} = {4.768}/{{\pi}/{4}}$

Dia = $\frac{6.07cm^2}{\sqrt{6.07}}$
$= 2.4cm^2 \approx 24\ mm$
So, this the limit so selection ram Dia to 24 mm or more then it

- **Annulus area over ram**

d1 = 63
d2 = 36

$(d1^2 - d2^2)\ {\pi}/{4}$
$(6.3^2 - 3.6^2)\ {\pi}/{4}$ *annuls area 20.99* cm^2
Annulus load = pressure × $load$
$= 130\ {kg}/{cm^2} \times 20.99cm^2$
= 2728.7 kg = 2.7287 T $\approx 3T$
To calculate forward load on ram and front
M32 × $2p = \emptyset = at\ root\ dia = 29.5$mm $\approx 2.95cm^2$

Root area = $2.95^2 \times {}^{\pi}/_{4}$

= 6.834 cm^2

- **Tensile / compression load**

Calculation over it

= Area × stear stress or compaction stress

= 6.834 cm^2 × 850 kg cm^2

= 5808.9 kg

Tensile = 5808.4 $>$ $then$ 4987.9

So, this is safer Dia of it

Ratio = 5809.6 ÷ 4987.9

= 1.16% more at actual load

4.5 Hand Calculations for CAE Validation

- Pressing Force (F): 5 tons = 49.050N
- Ram (Tie-Rod) Diameter = 36mm
- Tie-Rod = 4

Area of Tie-Rod

$$A = \frac{\pi}{4} \times d^2$$

$$A = \frac{\pi}{4} \times 36^2$$

$$A = 1017.87 mm^2$$

Stress in Tie-Rod (Tensile Strength)

$$\sigma = \frac{F}{A}$$

$$F = \frac{49050}{4}$$

$$F = 12262.5N$$

$$\sigma = \frac{12262.5}{1017.87}$$

$$\sigma = 12.047 \frac{N}{mm^2}$$

Factor of Safety

$$FOS = \frac{yield\ strength}{\max von\ mises\ stress}$$

$$FOS = \frac{250}{120}$$

$$FOS = 2.08$$

4.6 Design Validation Using HyperMesh (Static Analysis)

4.6.1 Geometry Cleanup and Meshing Part

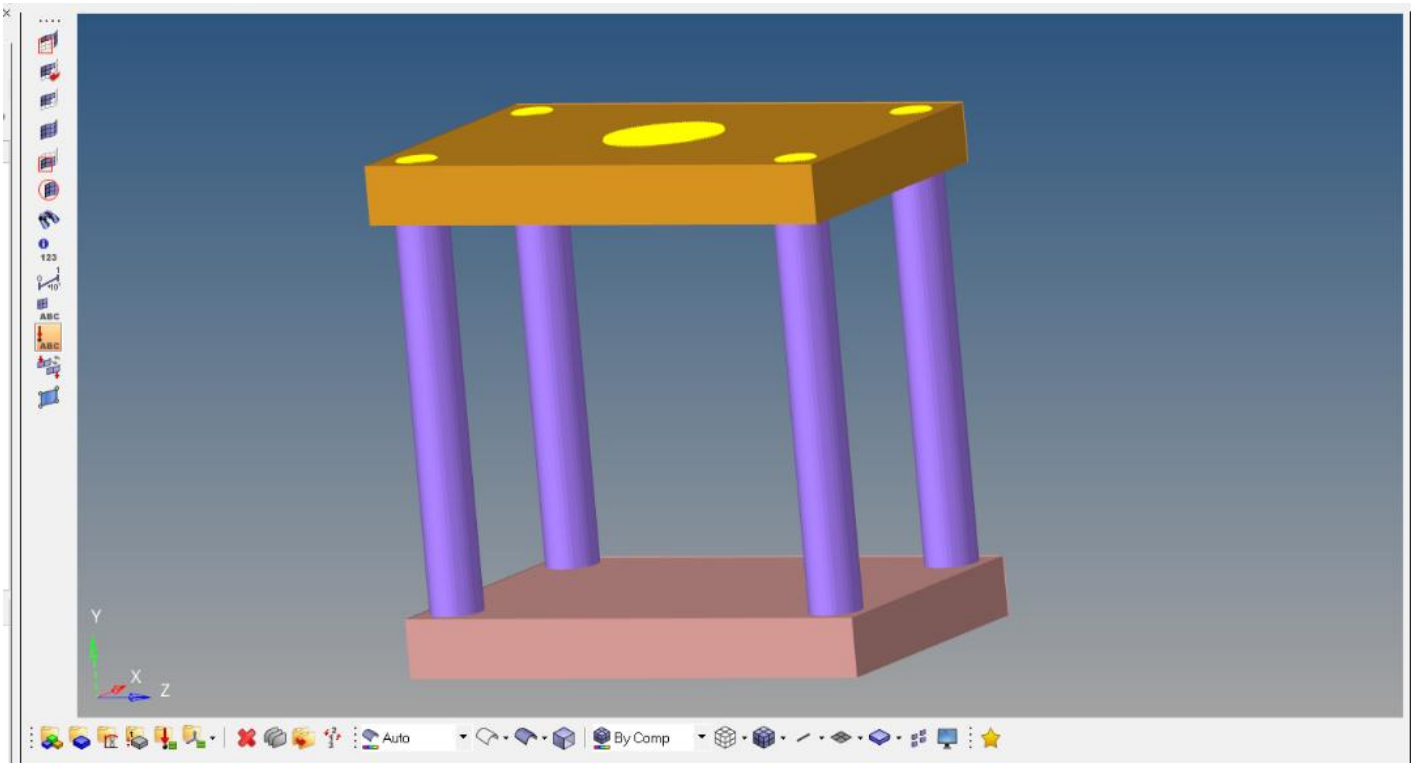


Fig 4.6.2 Solid Meshing of Assembly

Table 4.6.2 Assembly Solid Mesh Details (HyperMesh)

Sl. No.	Description	Specification
1	Model type	Assembly solid mesh
2	Software used	HyperMesh
3	Mesh type	3D solid mesh
4	Element type	Solid elements
5	Number of main components	6 (Top plate, bottom plate, 4 columns)
6	Meshing approach	Component-based solid meshing
7	Geometry source	CAD assembly
8	Mesh connectivity	Shared / connected nodes
9	Analysis purpose	Structural stress analysis

4.6.2 Force and Single Point Constraint

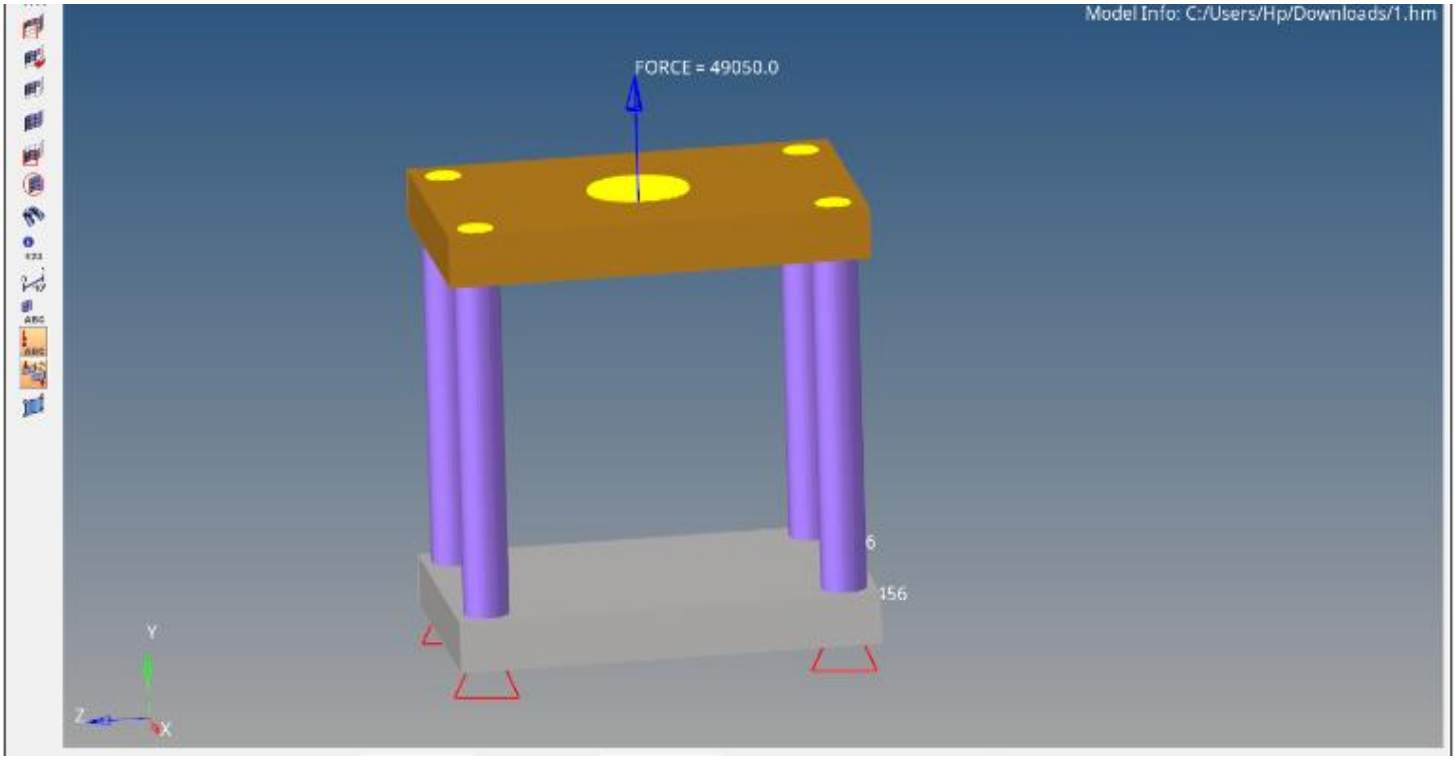


Fig 4.6.3 Force and Single Point Constraint

Table 4.6.3 Load and Boundary Condition Details

Sl. No.	Description	Specification
1	Analysis type	Static structural analysis
2	Software used	HyperMesh
3	Applied load type	Concentrated force
4	Magnitude of applied load	49,050 N
5	Direction of applied load	Vertical downward (–Y direction)
6	Location of applied load	Central hole on top plate
7	Support condition	Fixed support
8	Location of supports	Bottom plate (four corner supports)
9	Constraint type	All translational degrees of freedom constrained

4.7 Result

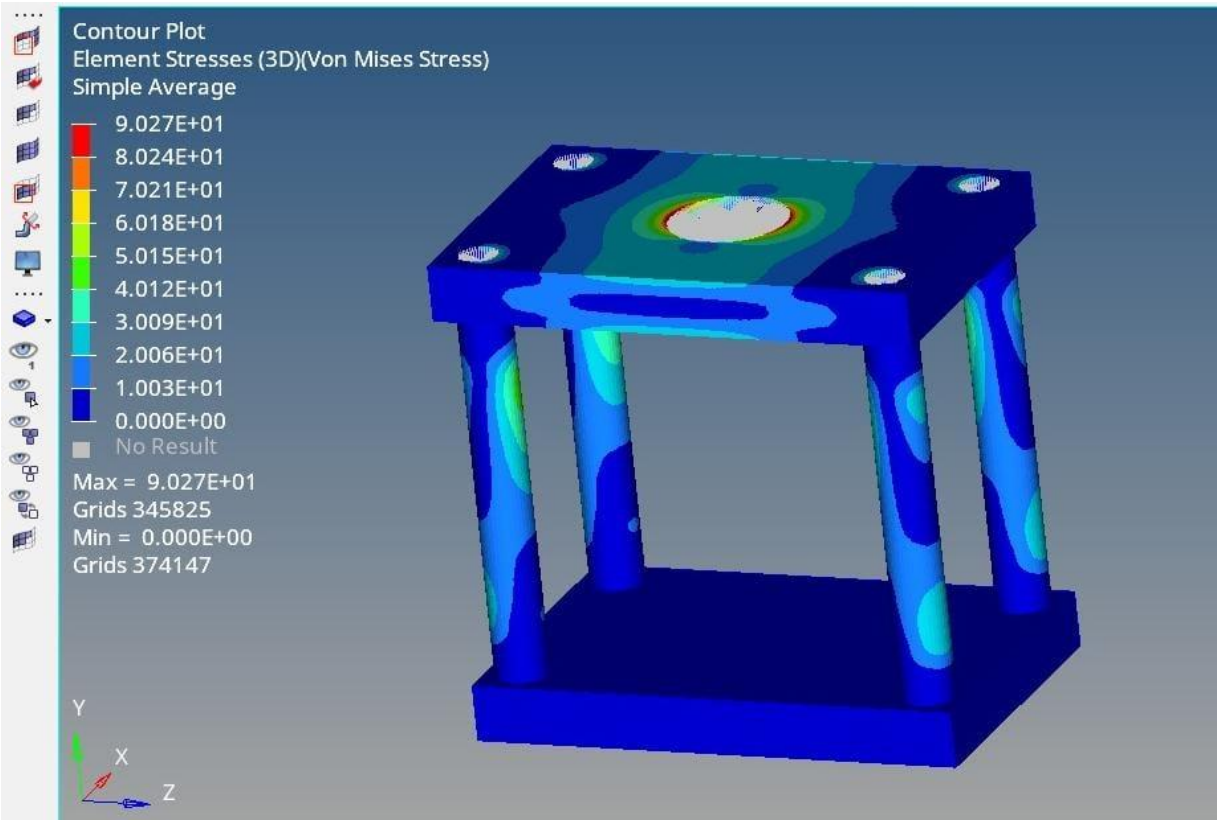


Fig 4.7 Stress Results of Hydraulic Press Machines

4.7.1 Comparison of Analytical and Simulation Results

Table 4.7.1 Comparison of Analytical and Simulation Results

Parameter	Analytical Result	Simulation Result (HyperMesh)
Total Applied Load (N)	49,050 N	49,050 N
Load per Column (N)	12,262.5 N	12,262.5 N
Area (mm²)	1,017.87	Based on actual geometry
Maximum Stress (MPa / N/mm²)	120 MPa	90.27 MPa
Yield Strength of Material (MPa)	250 MPa (Mild Steel)	250 MPa (Mild Steel)
Factor of Safety (FOS)	2.08	2.7
Method Used	Analytical ($\sigma = F / A$)	Finite Element Analysis

4.7.2 Software Part

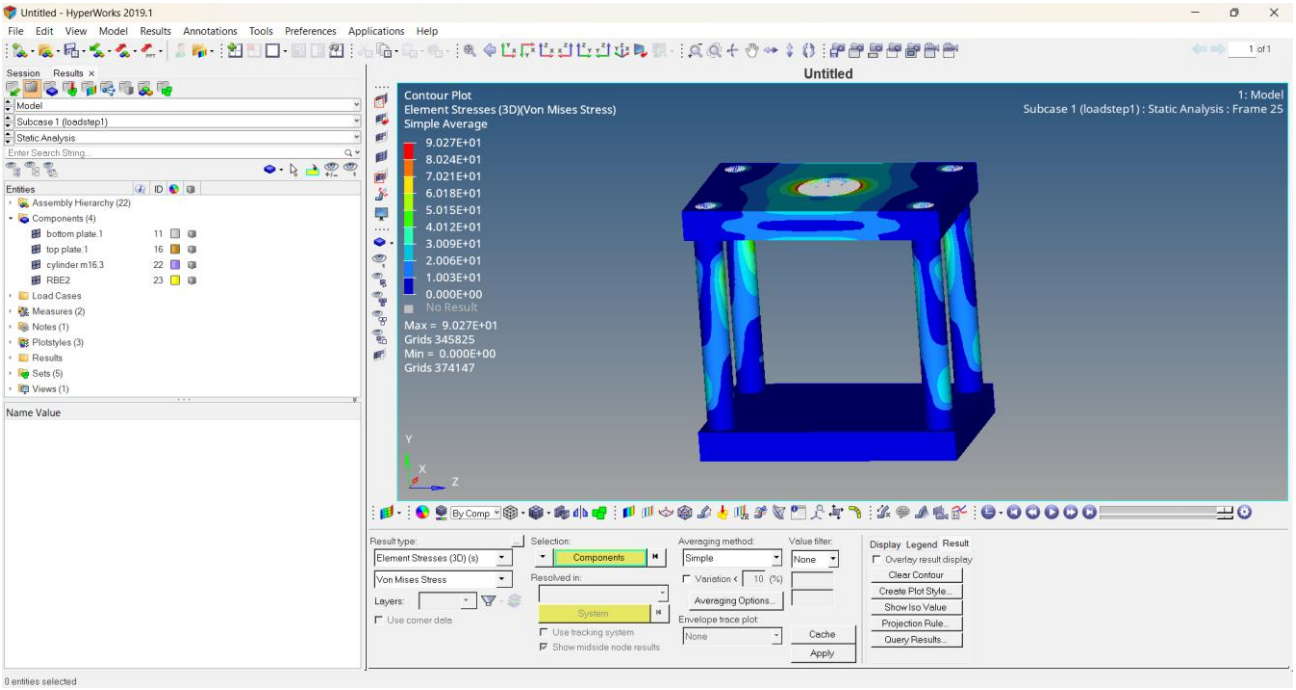


Fig 4.7.2 HyperMesh Software

HyperMesh is a high-performance finite element pre-processing and simulation software widely used in mechanical, automotive, aerospace, and manufacturing industries. It is primarily employed for geometry cleanup, meshing, and preparation of models for structural, thermal, and dynamic analysis. HyperMesh acts as a bridge between CAD models and finite element solvers by converting complex geometries into accurate and solver-ready finite element models. In engineering simulation, HyperMesh plays a crucial role in mesh generation, which directly affects the accuracy of analysis results. The software supports different types of elements such as 1D (beams), 2D (shells), and 3D (solid elements), allowing engineers to model real-world components with high precision. It provides extensive tools for controlling mesh quality parameters like element size, aspect ratio, skewness, and Jacobian, ensuring reliable and converged results. For structural analysis, HyperMesh enables the application of material properties, boundary conditions, and loads on the meshed model. It supports multiple solvers such as OptiStruct, ANSYS, Abaqus, and Nastran, making it highly versatile. Once the model is prepared, it can be exported to the desired solver for stress, strain, deformation, and factor of safety evaluation. By providing excellent control over mesh quality parameters such as aspect ratio, skewness, and Jacobian, HyperMesh improves the accuracy of stress and deformation results. In this project, HyperMesh is used to generate the finite element model, apply actual loading conditions, and perform stress analysis, and the obtained simulation results are compared with analytical calculations to validate the design and ensure structural safety.

4.8 COMPONENTS OF MACHINE



Fig 4.8.1 Double Acting Cylinder

A double-acting cylinder is a hydraulic or pneumatic actuator in which fluid pressure is applied alternately to both sides of the piston to produce motion in both forward and return directions. In its working, pressurized fluid enters one side of the cylinder to extend the piston rod and perform useful work, and then fluid is supplied to the opposite side to retract the piston, ensuring controlled movement in both strokes. Unlike single-acting cylinders, it does not rely on springs or external forces for return motion. Double-acting cylinders are widely used in hydraulic presses, powder metallurgy compaction, machine tools, lifting systems, and automation equipment, where precise control, uniform force application, and reliable bidirectional movement are required.



Fig 4.8.2 Upper Plate



Fig 4.8.3 Lower Plate

- Upper Plate (Upper Platen): The moving plate connected to the piston rod that transmits hydraulic force to the die or workpiece during pressing.

- Lower Plate (Lower Platen): The fixed base plate that supports the die and workpiece and withstands the reaction force during operation.



Fig 4.8.4 Piston Rod

The piston rod is a strong metal rod connected to the piston inside a hydraulic cylinder that transmits force and motion to the press components. It moves linearly under hydraulic pressure and is commonly made of hardened, chrome-plated steel for strength and wear resistance.



Fig 4.8.5 Steel Stand Frame

The steel stand frame is the main supporting structure of a hydraulic machine that holds all components in alignment and safely carries the applied load. It is made from mild or structural steel to provide strength, rigidity, and stability during operation.



Fig 4.8.6 Tie-Rod

A tie rod in a hydraulic machine is a high-strength threaded steel rod used to clamp and hold the press plates or frame together**.** It ensures proper alignment, rigidity, and safe load distribution during pressing operations.



Fig 4.8.7 Seal Box & Flange

- Seal Box: The seal box (gland) houses sealing elements that prevent hydraulic oil leakage and protect the piston rod during movement.

- Flange: The flange is a mounting and support component used to secure the cylinder or seal box to the machine frame, ensuring proper alignment and strength.

4.9 Magnesium Powder

Magnesium powder is a finely divided metallic form of magnesium, known for its low density, high strength-to-weight ratio, and high chemical reactivity. It appears as a silver-grey powder and is widely used in industries such as aerospace, automotive, pyrotechnics, metallurgy, and chemical processing. Due to its lightweight nature (density ≈ 1.74 g/cm³), magnesium is one of the lightest structural metals, making it highly suitable for applications where weight reduction is critical. From a material science perspective, magnesium powder exhibits excellent compressibility, which makes it suitable for powder metallurgy and pelletization processes. However, in loose powder form, magnesium poses several challenges such as poor flowability, oxidation, dust formation, and high risk of ignition when exposed to heat, sparks, or friction. These issues make handling, storage, and transportation of magnesium powder unsafe and inefficient. Therefore, converting magnesium powder into compact pellets is essential to improve safety and usability. In the context of this project, magnesium powder is compacted using a hydraulic press machine to form dense and uniform pellets. Hydraulic compaction applies controlled and uniform pressure, which increases particle bonding and reduces porosity within the pellet. Pelletization improves mechanical strength, dimensional stability, and handling safety, while also minimizing material loss and contamination. The use of hydraulic pressure is particularly advantageous because it provides smooth force application, reducing the risk of sudden energy release that could trigger ignition. Magnesium powder pellets produced through this process are easier to store, transport, and use in downstream applications such as alloy preparation, combustion-based processes, and experimental research. Additionally, pelletization reduces the exposed surface area of magnesium, thereby lowering oxidation rates and improving shelf life. Thus, magnesium powder serves as a critical raw material in this project, and its conversion into pellets using a hydraulic press significantly enhances both operational safety and industrial applicability.



Fig 4.9.1 Magnesium Powder



Fig 4.9.2 Pellets from Magnesium Powder

4.9.3 Material Properties of Magnesium Powder

Table 4.9.3 Material Properties of Magnesium Powder

Property	**Value**	**Unit**
Chemical Symbo	Mg	–
Atomic Weight	Mg	g/mol
Density	1.74	g/cm³
Melting Point	650	°C
Boiling Point	1,091	°C
Elastic Modulus	45	GPa
Poisson's Ratio	0.29	–
Compressibility	Good (suitable for pelletization)	–

4.9.4 MATERIAL SELECTION FOR HYDRAULIC PRESS MACHINE

Table 4.9.4 Material Selection for Hydraulic Press Machine

COMPONENT	MATERIAL	REASON
Frame (H-Frame)	Mild Steel (IS 2062)	High structural strength, good machinability, cost-effective.
Hydraulic Cylinder	Seamless Steel Tube	Withstands internal pressure, high fatigue resistance.
Ram / Piston Rod	EN 24 (Alloy Steel, hardened and ground)	High tensile strength and wear resistance. Chrome-plated for longevity.
Upper & Lower Platens	MS Plate (IS 2062 E250)	Strong under compression, easy to machine flat surfaces.
Die/Mold Block	Tool Steel (H13 / D2)	High hardness and heat resistance; ideal for powder compaction molds.
Base Plate	Mild Steel (IS 2062)	Needs to support total load; economical and strong.
Guide Pillars	EN 8 or EN 24, hardened	Wear resistance and dimensional stability under motion.

CHAPTER 5 COST EXPENDITURE

Table 5.1 Cost Expenditure

component	Diameter (mm)	Inner Diameter (mm)	Length (mm)	Others	Amount (Rupees)
Cylinder	63	50	230		1290/-
Flange	105	80	12		378/-
Ram	40		310		859/-
Piston Head	60		60		421/-
Seal Box	80	80	75		967/-
END Cover	80	80	50		720/-
END Cover NUT	160		0		
Wiper Ring	160		0		
Rod Seal	120		0		
Piston Seal					67/-
O-Ring					150/-
Turcite Rod					450/-
Painting					75/-
Hardware					700/-
Assy/Test					1700/-
Top Plate	350	210	50	1	5672/-
Support Plate	300	100	16		
Moving Plate	200	200	25	1	1542/-
Bottom Plate	350	210	50	1	5672/-
Heater	10				
Thermo Couple	10		200		
Tie-Rod	361		325	4	5952/-
Guide Bush	80	80	75		
Split Flange	80	80	30		
Turcite	50		157.65		
MS Plate	450	200	8	1	1264/-
Angle	70	10	750	4	3687/-
Support Plate	70	10	500	4	2458/-
Booster Plate	300	100	16		
	260	260	32		
				Total	34,020/-

CHAPTER 6

RESULTS AND DISCUSSION

The design and development of the hydraulic press machine for making pellets from magnesium powder was successfully completed, and the system performed as intended during testing. The developed hydraulic press was able to generate sufficient and controlled compressive force required for compacting magnesium powder into solid pellets. The pellets produced were uniform in shape, consistent in dimensions, and exhibited good mechanical integrity, indicating effective pressure transmission and proper die–punch alignment. The hydraulic system operated smoothly without pressure fluctuations, ensuring repeatability and reliability in pellet formation. From a structural perspective, the press frame and major components safely withstood the applied loads during operation. No visible deformation, cracks, or instability were observed, confirming the validity of the design calculations and appropriate material selection. The uniform distribution of stress across the structure highlights the robustness of the machine and its suitability for repeated compaction cycles. The controlled hydraulic action also minimized shock loading, which is essential for long-term durability of the press components.



Fig 6.1 Hydraulic Press Machine

CHAPTER 7
APPLICATIONS AND ADVANTAGES

Applications

1. Powder Metallurgy Industry
   Used for compacting magnesium powder into uniform green pellets before sintering, ensuring controlled density and shape.
2. Pyrotechnics and Defense Applications
   Magnesium pellets are widely used in flares, igniters, and incendiary compositions where controlled size and density are critical.
3. Chemical and Metallurgical Research
   Facilitates safe handling and testing of magnesium by converting fine powder into compact pellets for laboratory experiments.
4. Automotive and Aerospace Components
   Magnesium-based pellets can be further processed into lightweight components, contributing to weight reduction and fuel efficiency.
5. Energy and Battery Applications
   Used in research and development of magnesium-based energy storage systems and thermal reaction studies.
6. Safe Storage and Transportation
   Pelletizing magnesium powder reduces dust formation, making storage and transport safer and more manageable.

Advantages

1. High Compaction Force and Uniform Density
   Hydraulic presses provide consistent and controllable pressure, resulting in pellets with uniform density and improved mechanical strength.
2. Improved Safety
   Reduces the risk of fire and explosion associated with handling loose magnesium powder by converting it into compact pellets.
3. Better Dimensional Accuracy
   Produces pellets with precise size and shape, suitable for automated handling and further processing.
4. Cost-Effective Operation
   Simple construction, low maintenance, and efficient power usage make the hydraulic press economical for small- to medium-scale production.
5. Process Flexibility
   Pressure, dwell time, and pellet dimensions can be easily adjusted to meet different application requirements.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

CONCLUSION

The design and development of the hydraulic press machine for making pellets from magnesium powder has been successfully completed and tested. The developed system effectively converts loose magnesium powder into compact, uniform, and mechanically stable pellets using controlled hydraulic pressure. The machine demonstrated smooth operation, reliable force generation, and structural stability under working conditions, validating the design calculations and material selection. The produced pellets showed consistent dimensions and improved density, making them safer to handle, store, and transport compared to lose magnesium powder.

The hydraulic press also enhanced operational safety by reducing direct manual handling of magnesium powder, which is highly reactive in nature. The simple construction, cost-effective design, and ease of operation make the developed machine suitable for laboratory use as well as small-scale industrial applications. Overall, the project meets its stated objectives and proves the feasibility of using a hydraulic press for magnesium powder pelletization in powder metallurgy processes.

FUTURE WORK – PLC INTEGRATION

The performance and efficiency of the hydraulic press can be significantly enhanced by integrating a Programmable Logic Controller (PLC) system. PLC-based automation would enable precise control over pressing parameters such as pressure, stroke length, dwell time, and cycle speed. This will ensure improved consistency in pellet density and reduce operator dependency.

In future development, pressure sensors and linear position sensors can be interfaced with the PLC to provide real-time feedback and closed-loop control. A Human-Machine Interface (HMI) can be added for easy parameter setting, monitoring, and fault diagnosis. Safety features such as emergency stop, overload protection, and interlocking systems can also be implemented through PLC logic, further improving operator safety. Additionally, PLC integration will allow automatic sequencing of operations, making the machine suitable for continuous and large-scale production. Data logging and process monitoring can be used for quality control and performance analysis. With these enhancements, the hydraulic press can be upgraded into a fully automated, reliable, and industry-ready system for magnesium powder pellet manufacturing.

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